Rahul Pillalamarri

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Education

- **Texas A&M University** *Aug. 2024 - Present*
  M.S. in Computer Science, Concentration in Machine Learning/AI (In Progress) GPA: 3.50
- **University of Washington** *Sept.2021 - Jun. 2023*
  B.S. in Computer Science, Minor in Mathematics GPA: 3.51

Technical Skills

Languages: Python, C/C++, C#, JavaScript, Java, MATLAB, SQL, Kotlin, Bash, XML
Technologies: Pytorch, TensorFlow, sklearn, CUDA, OpenCV, AWS, Azure, Spark, Hadoop, React, Android Studio
Expertise: ML/AI, Deep Learning, NLP, Big Data, Real-Time Embedded Systems, High Performance Computing

Experience

- **Deep Learning Fundamentals Lab** *June 2025 - Present*
  Graduate Researcher (under Dr. Tomer Galanti - Texas A&M CSE) College Station, TX
  - **Directional Geometric Loss for Robust Fine-Tuning:** Designed a loss function grounded in neural-collapse and contrastive-learning theory to fine-tune CLIP, mitigating catastrophic forgetting while achieving comparable downstream performance on ImageNet-1K
  - **Comparative Study of Reasoning Models:** Evaluated the Hierarchical Reasoning Model (HRM) against state-of-the-art LLM baselines (Mistral, BLOOM, Llama) on pattern recognition and graph-reasoning tasks.
  - **High Performance Computing:** Wrote multi-threaded, multi-GPU training pipelines in Python to run large-scale experiments on TAMU's Grace A100 cluster and cloud H100 servers.
- **Boeing** *Sept. 2023 - Present*
  Software Engineer (Secret Security Clearance) St. Louis, MO
  - **Real-Time Embedded Systems:** Implemented integral services to enable the autonomous flight control of the MQ25 refueller in C++ using multi-threading and live data streams at the bit level; this excludes additional classified work.
  - **Testing & Validation:** Developed scripts to accelerate operational flight program validation in Hardware-in-the-Loop (HIL) environments, improving testing speed by 60% and supporting end-of-sprint release cycles.
  - **Image Analysis:** Created a test pipeline leveraging OpenCV-based image analysis to validate flight systems.
- **MCG Health** *Jun. 2023 - Sept. 2023*
  Software Engineer Intern Seattle, WA
  - **API Development:** Designed and deployed APIs in C# on Azure to speed up the authoring and testing of XML evidence-based healthcare guidelines by 30%.
  - **API Testing:** Wrote unit tests using Moq and Fluent-Assertions libraries for micro-service testing.
- **Sensoria Health** *Aug. 2021 - Oct. 2022*
  Software Engineer Intern Redmond, WA
  - **Android Development:** Built two mobile applications in Java and Kotlin to configure and gather data from company health-tech devices to bridge the physician and patient gap during patient rehabilitation.
  - **Cloud Systems:** Integrated Google Cloud Platform Firebase into mobile applications to implement data pipelines to transmit patient recovery data to physicians, onboard patients, and test product.

Projects

Generative Models (WGAN): Implemented the Wasserstein GAN as described by Arjovsky et al. in the 2017 paper, reproducing their results. Extended the model with gradient penalty to further stabilize training.

Computer Vision: Developed a custom message passing system to encrypt/decrypt secret messages stored in color boxes from real-world images using OpenCV.

NLP: Created an interactive NLP-driven web app that leverages speech-to-text libraries to analyze user input and provide pronunciation and grammar feedback, integrated into a JavaScript front-end.

Text Generation: Built a hybrid CNN LSTM model in PyTorch to generate text from Reddit comments.

Extracurriculars + Awards

Extracurriculars: CSE Tutor at UW Quantitative Success Center, Chess Instructor, Private Math Tutor, long-distance runner, mountaineer, video editor
Awards: UWB Hacks the Cloud Hackathon Winner (2022), Deans List (2021 - 2023)